

Zihan Zhou

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Education

- 2023 – now  **Ph.D. Physics, Princeton University**
Thesis advisor: Matias Zaldarriaga
- 2021 – 2023  **M.A. Physics, Princeton University**
- 2017 – 2021  **S.Bc. Physics, University of Science and Technology of China (USTC)**
Thesis title: *Searching for beyond standard model physics with gravitational waves.*
Thesis advisor: Yi-Fu Cai

Research Interest

My research initially centered on gravitational wave astrophysics in the strong-field regime, with particular emphasis on effective field theory, scattering amplitudes, and precision waveform modeling for tidal effects, relativistic binaries, and resummation in gravitational waves. More recently, my interests have broadened to include quantum dynamics and AI for physics, with particular emphasis on the interplay between dynamical observables, symmetry, and nonequilibrium behavior in quantum many-body systems, as well as agent frameworks for astrophysical inference and discovery in fundamental physics.

Quantum Dynamics and AI for Physics

I work on dynamical observables, symmetry, and nonequilibrium behavior in quantum many-body systems, including a bootstrap framework that reconstructs hidden finite-group symmetries from spectral correlations, and a mechanism for quantum Hilbert space fragmentation through entangled frozen states. In parallel, I develop and explore agent frameworks for cross-disciplinary scientific reasoning across gravitational-wave astronomy, cosmology, and effective field theory, and have used AI agent as a research collaborator to discover a closed-form polytope formula (“Waterhedron”) for n -wave amplitudes in classical fluid scattering.

Gravity, Compact Objects, and Gravitational Waves

I study compact objects and gravitational waves using effective field theory, scattering amplitudes, and precision waveform modeling. This includes relativistic binary dynamics, tidal effects, and resummation methods in strong gravity, as well as data analysis strategies for current and future detectors, including machine-learning approaches to signal modeling and search.

Talks

- Invited Talks *Amplitudes, Strong-Field Gravity and Resummation* (virtual), Apr. 13, 2026
 High Energy Theory Group Meeting, Columbia, New York, Feb. 25, 2026
 Theoretical Physics Seminar, KITS, Beijing, Jan. 14, 2026
 IAS Physics+Astro Meeting 2, IAS, Princeton, Dec. 15, 2025
 IAS Physics+Astro Meeting 1, IAS, Princeton, Dec. 9, 2025
 Probing New Physics with Accelerators and the Cosmos (virtual), Oct. 28, 2025
 Workshop on QNM and BHPT, IBS, Daejeon, Korea, May. 28, 2025
 IAS Amplitudes Group Meeting, IAS, Princeton, Apr. 15, 2025
 High Energy Theory Seminar, Fudan, Shanghai, Dec. 24, 2024
 QCD Meets Gravity 2024, NTU, Taiwan, Dec. 12, 2024
 High Energy Theory Informal Discussion, Caltech, Pasadena, Dec 3, 2024

Talks (continued)

Scattering Amplitudes and Gravity in East Asia (virtual), May. 7, 2024
IAS Physics Group Meeting, IAS, Princeton, May. 1, 2024
The S-Matrix Marathon, IAS, Princeton, Mar. 21, 2024
IAS Cosmology Lunch, IAS, Princeton, Mar. 18, 2024
High Energy Theory Seminar, UBC, Vancouver, Feb. 12, 2024
High Energy Theory Seminar, CMU, Pittsburgh, Oct. 27, 2023
High Energy Theory Seminar, Columbia, New York, Sep. 25, 2023
Strong Gravity Seminar, PI, Waterloo, Jun. 15, 2023
High Energy Theory Seminar, UPenn, Philadelphia, Feb. 22, 2023
WQFT Seminar (virtual), Jan. 23, 2023
Joint HEP-TH Seminar (virtual), Nov. 30, 2022
High Energy Informal Discussion, Caltech, Pasadena, Oct. 26, 2022

Contributed Talks *APS March Meeting*, Denver, Mar. 19, 2026
 APS March Meeting, Anaheim, Mar. 17, 2025
 Strong Gravity Group Meeting, NBI, Copenhagen, Oct. 5, 2023
 APS April Meeting, New York, Mar. 28, 2022
 IAS Astro Coffee, IAS, Princeton, Jan, 2022
 Theoretical Physics Seminar, ITP, Beijing, May. 28, 2021
 Theoretical Physics Seminar, ITP, Beijing, Nov, 2020
 Workshop on Dark Matter and Gravitational Waves, PMO, Nanjing, July, 2020

Mentoring, Teaching and Reviewing

Mentoring Students

2025 **Liam Wang**, one 1st author paper in prep
2024 **Joshua Shterenberg**, one 1st author paper

Teaching

Princeton	Teaching Assistant, Physics for Future Leaders (PHY115)	Fall 2024
	Teaching Assistant, Physics for the Life Sciences (PHY108)	Spring 2024
	Teaching Assistant, General Physics I (PHY103)	Fall 2023
	Teaching Assistant, Advanced Physics (Electromagnetism) (PHY106)	Spring 2023
	Teaching Assistant, Introductory Physics I (PHY101)	Fall 2022
USTC	Teaching Assistant, Statistical Physics	Spring 2021
	Teaching Assistant, Electrodynamics	Spring 2020
	Teaching Assistant, Complex Analysis	Fall 2019

Awards

Princeton	Jane Street Graduate Research Fellowship	2025
USTC	Yonghuai Guo Award	2021
	Outstanding Undergraduate Thesis Award	2021
	Meritorious Winner in 2020 Interdisciplinary Contest in Modeling	2020
	Second Prize in National Mathematical Contest in Modeling	2019
	9th Place (Anhui), National Mathematics Competition for College Students	2018

Selected Research Publications

Journal Articles

- 1 C. Bai, Z. Zhou, B. Lapierre, and S. Ryu, “Bootstrapping Symmetries in Quantum Many-Body Systems from the Cross Spectral Form Factor,” Apr. 2026. arXiv: 2604.01296 [quant-ph].
- 2 C.-H. Chang, C.-H. Shen, and Z. Zhou, “Gravitational Sommerfeld Effects: Formalism, Renormalization, and Perturbation to $O(G^{10})$,” Apr. 2026. arXiv: 2604.14112 [hep-th].
- 3 M. M. Ivanov, Y.-Z. Li, J. Parra-Martinez, and Z. Zhou, “Gravitational Raman Scattering: a Systematic Toolkit for Tidal Effects in General Relativity,” Feb. 2026. arXiv: 2602.06951 [hep-th].
- 4 Z. Zhou, T.-H. Yang, and B.-T. Chen, “Quantum Hilbert Space Fragmentation and Entangled Frozen States,” Apr. 2026. arXiv: 2604.05218 [quant-ph].
- 5 S. Akhtar, Y. F. Bautista, C. Iossa, and Z. Zhou, “Five-dimensional gravitational Raman scattering: Scalar wave perturbations in Schwarzschild-Tangherlini spacetime,” *Phys. Rev. D*, vol. 112, no. 8, p. 085018, 2025.  DOI: 10.1103/rglg-kfxq. arXiv: 2505.21489 [hep-th].
- 6 G. Cabass, D. Wadekar, M. Zaldarriaga, and Z. Zhou, “Generating optimal Gravitational-Wave template banks with metric-preserving autoencoders,” Nov. 2025. arXiv: 2511.10466 [gr-qc].
- 7 H. S. Chia, Z. Zhou, and M. M. Ivanov, “Tidal heating constraints for black holes and exotic compact objects from the LIGO-Virgo-KAGRA data,” *Phys. Rev. D*, vol. 111, no. 6, p. 063002, 2025.  DOI: 10.1103/PhysRevD.111.063002. arXiv: 2404.14641 [gr-qc].
- 8 M. Correia, H. S. Hannesdottir, G. Isabella, A. M. Wolz, and Z. Zhou, “Gravitational Physics from Scattering Amplitudes,” *Lect. Notes Phys.*, vol. 1041, pp. 259–308, 2025.  DOI: 10.1007/978-3-031-90352-6_7.
- 9 M. M. Ivanov, Y.-Z. Li, J. Parra-Martinez, and Z. Zhou, “Resummation of Universal Tails in Gravitational Waveforms,” *Phys. Rev. Lett.*, vol. 135, no. 14, p. 141401, 2025.  DOI: 10.1103/jzdd1-qzkt. arXiv: 2504.07862 [hep-th].
- 10 J. Shterenberg and Z. Zhou, “Fisher forecast of finite-size effects with future gravitational wave detectors,” *Phys. Rev. D*, vol. 111, no. 8, p. 084068, 2025.  DOI: 10.1103/PhysRevD.111.084068. arXiv: 2410.00294 [gr-qc].
- 11 Z. Zhou, G. M. Tomaselli, I. Martínez-Rodríguez, and J. Li, “Modeling Tidal Disruptions with Dynamical Tides,” *Astrophys. J.*, vol. 990, no. 2, p. 198, 2025.  DOI: 10.3847/1538-4357/adf190. arXiv: 2504.16025 [astro-ph.HE].
- 12 Y. F. Bautista, G. Bonelli, C. Iossa, A. Tanzini, and Z. Zhou, “Black hole perturbation theory meets CFT2: Kerr-Compton amplitudes from Nekrasov-Shatashvili functions,” *Phys. Rev. D*, vol. 109, no. 8, p. 084071, 2024.  DOI: 10.1103/PhysRevD.109.084071. arXiv: 2312.05965 [hep-th].

- 13 M. M. Ivanov, Y.-Z. Li, J. Parra-Martinez, and Z. Zhou, “Gravitational Raman Scattering in Effective Field Theory: A Scalar Tidal Matching at $\mathcal{O}(G_3)$,” *Phys. Rev. Lett.*, vol. 132, no. 13, p. 131401, 2024.  DOI: 10.1103/PhysRevLett.132.131401. arXiv: 2401.08752 [hep-th].
- 14 M. V. S. Saketh, Z. Zhou, S. Ghosh, J. Steinhoff, and D. Chatterjee, “Investigating tidal heating in neutron stars via gravitational Raman scattering,” *Phys. Rev. D*, vol. 110, p. 103001, 2024.  DOI: 10.1103/PhysRevD.110.103001. arXiv: 2407.08327 [gr-qc].
- 15 M. V. S. Saketh, Z. Zhou, and M. M. Ivanov, “Dynamical tidal response of Kerr black holes from scattering amplitudes,” *Phys. Rev. D*, vol. 109, no. 6, p. 064058, 2024.  DOI: 10.1103/PhysRevD.109.064058. arXiv: 2307.10391 [hep-th].
- 16 M. M. Ivanov and Z. Zhou, “Revisiting the matching of black hole tidal responses: A systematic study of relativistic and logarithmic corrections,” *Phys. Rev. D*, vol. 107, no. 8, p. 084030, 2023.  DOI: 10.1103/PhysRevD.107.084030. arXiv: 2208.08459 [hep-th].
- 17 M. M. Ivanov and Z. Zhou, “Vanishing of Black Hole Tidal Love Numbers from Scattering Amplitudes,” *Phys. Rev. Lett.*, vol. 130, no. 9, p. 091403, 2023.  DOI: 10.1103/PhysRevLett.130.091403. arXiv: 2209.14324 [hep-th].
- 18 Y.-F. Cai, X.-H. Ma, M. Sasaki, D.-G. Wang, and Z. Zhou, “Highly non-Gaussian tails and primordial black holes from single-field inflation,” *JCAP*, vol. 12, p. 034, 2022.  DOI: 10.1088/1475-7516/2022/12/034. arXiv: 2207.11910 [astro-ph.CO].
- 19 Y.-F. Cai, X.-H. Ma, M. Sasaki, D.-G. Wang, and Z. Zhou, “One small step for an inflaton, one giant leap for inflation: A novel non-Gaussian tail and primordial black holes,” *Phys. Lett. B*, vol. 834, p. 137461, 2022.  DOI: 10.1016/j.physletb.2022.137461. arXiv: 2112.13836 [astro-ph.CO].
- 20 Y. Chen, C. Li, Y. Mizuno, *et al.*, “Birefringence tomography for axion cloud,” *JCAP*, vol. 09, p. 073, 2022.  DOI: 10.1088/1475-7516/2022/09/073. arXiv: 2208.05724 [hep-ph].
- 21 S. Kotecha *et al.*, “Cosmic filaments delay quenching inside clusters,” *Mon. Not. Roy. Astron. Soc.*, vol. 512, no. 1, pp. 926–944, 2022.  DOI: 10.1093/mnras/stac300. arXiv: 2110.13419 [astro-ph.GA].
- 22 Y.-F. Cai, J. Jiang, M. Sasaki, V. Vardanyan, and Z. Zhou, “Beating the Lyth Bound by Parametric Resonance during Inflation,” *Phys. Rev. Lett.*, vol. 127, no. 25, p. 251301, 2021.  DOI: 10.1103/PhysRevLett.127.251301. arXiv: 2105.12554 [astro-ph.CO].
- 23 Z. Zhou, J. Yan, A. Addazi, Y.-F. Cai, A. Marciano, and R. Pasechnik, “Probing new physics with multi-vacua quantum tunnelings beyond standard model through gravitational waves,” *Phys. Lett. B*, vol. 812, p. 136026, 2021.  DOI: 10.1016/j.physletb.2020.136026. arXiv: 2003.13244 [astro-ph.CO].
- 24 Z. Zhou, J. Jiang, Y.-F. Cai, M. Sasaki, and S. Pi, “Primordial black holes and gravitational waves from resonant amplification during inflation,” *Phys. Rev. D*, vol. 102, no. 10, p. 103527, 2020.  DOI: 10.1103/PhysRevD.102.103527. arXiv: 2010.03537 [astro-ph.CO].